

Literature Survey on Automated and AI-Based Inspection Techniques for PCBAs

This literature survey is based on industry standards, company documentation, and academic research papers. It focuses on conventional and AI-based inspection techniques used for Printed Circuit Board Assemblies (PCBAs).

1. Introduction

Printed Circuit Boards (PCBs) and Printed Circuit Board Assemblies (PCBAs) are essential components in modern electronic products. As electronic systems become more complex and compact, ensuring defect-free PCBAs has become a critical challenge. Manual inspection is time-consuming and unreliable, leading to the adoption of automated inspection and testing methods.

2. Industry-Standard PCBA Testing Methods (Averna Test Systems)

2.1 Automated Optical Inspection (AOI)

AOI uses high-resolution cameras, advanced lighting systems, and AI-powered vision software to detect visible defects such as missing components, soldering faults, and alignment issues. Both 2D and 3D AOI systems ensure quality, consistency, and repeatability in manufacturing.

2.2 X-Ray Inspection

X-ray inspection allows visualization of internal layers and hidden solder joints such as BGAs. It helps identify defects like solder voids and internal shorts that cannot be detected optically.

2.3 In-Circuit Testing (ICT)

ICT detects electrical defects including shorts, opens, and incorrect component values without powering the PCB. This method requires dedicated test points and fixtures, which are increasingly difficult to implement due to high PCB density.

2.4 Fly Probe Testing

Fly probe testing uses movable probes instead of fixed fixtures, making it ideal for prototypes and low-volume production. Although slower than ICT, it is cost-effective and flexible.

2.5 Functional and Burn-in Testing

Functional testing verifies PCBA operation under real-world conditions, while burn-in testing stresses the board under extreme conditions to identify early-life failures and ensure long-term reliability.

3. Academic Research on PCB Defect Detection (IEEE Paper)

The IEEE research paper (Document ID: 10200565) highlights the limitations of manual PCB inspection and emphasizes the need for automated defect detection systems. Defects such as shorts, pinholes, spurs, and mouse bites can lead to performance degradation and system failure. Automated image-based inspection systems improve accuracy, reduce human dependency, and

enhance manufacturing efficiency.

4. AI-Based Industrial PCB Inspection (SwitchOn DeepInspect®)

DeepInspect® is an AI-powered visual inspection software capable of detecting surface defects, performing OCR/OCV checks, dimensional analysis, and component presence verification. The system achieves up to 99.5% accuracy using fewer than 200 training images and integrates automated rejection mechanisms for faulty PCBAs.

5. Comparative Analysis

Conventional inspection methods remain essential but face limitations with increasing PCB complexity. AI-powered systems offer higher accuracy, adaptability, and reduced inspection time, making them suitable for modern electronics manufacturing environments.

6. Conclusion

The literature survey concludes that integrating AI-based inspection systems with conventional testing methods significantly enhances PCBA quality control. Future PCB manufacturing will increasingly rely on intelligent automated inspection solutions to meet quality and reliability demands.

References

- Avera Test Systems – Common PCBA Testing Methods
- IEEE Research Paper: Automated PCB Defect Detection (Document ID: 10200565)
- SwitchOn – DeepInspect® AI-Based PCB Inspection Software